ISAAC SHEARD

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Education

B.S. Aerospace Engineering **Expected May 2028**

Massachusetts Institute of Technology (MIT) - Cambridge, MA | GPA: 4.3/5.0

- **Relevant Coursework:** Advanced Machining; Materials & Structures; Signals & Systems; Aerospace Probability & Statistics; Autonomous Machines; Thermodynamics & Prop.; Fluid Dynamics; Controls; Aerodynamics (IP); Feedback Control Systems (IP); Dynamics (IP);
- **Programs:** Gordon Engineering Leadership (GEL), Undergraduate Practice Opportunities Program (UPOP - Completed 2026)

Professional & Research Experience

Skyrun AI – San Francisco, CA

Robotics Engineering Intern (Summer 2026) | Contract Engineer (Current) *May 2026 - Present*

- Designed and built 2 autonomous rovers from scratch, owning mechanical, electrical, compute, sensor, and autonomy integration; took the second platform from chassis delivery to first autonomous SLAM map in 9 days and field-proven autonomous navigation in 6 weeks
- Built the sim-to-real infrastructure to deploy autonomy across 2 mechanically different rovers and simulation, enabling 6 autonomous behaviors to transfer between platforms without rewriting the underlying autonomy logic
- Entrusted with technical hiring, onboarding, and core robotics tooling, conducting interviews for master's-level engineering candidates, training new hires on the rover/autonomy stack, and owning core robotics tooling

MIT Hypersonics Research Lab — Cambridge, MA

Hypersonics Researcher *September 2026 – Present*

- Contributing to a hypersonic reentry plasma modeling project aimed at predicting electron density and radio-communication blackouts around the OREX spacecraft using US3D CFD
- Reconstructed OREX vehicle geometry from NASA/JAXA technical drawings and verified dimensions against published data; developing a structured axisymmetric CFD grid in Pointwise for US3D simulations

AGCO **– Jackson, MN**

Embedded Software Engineering Intern *Jun 2024 - Aug 2024; Jun 2025 - Aug 2025*

- Led the design and implementation of a user interface and CAN signal routing for an integrated weather station system on the sprayer platforms
- Achieved up to 200x faster runtimes across four optimization projects and built calibration software for the RoGator sprayer tractor

MIT Media Lab, Space Enabled – Cambridge, MA

Zero Robotics Researcher *Jan 2025 - Jan 2026*

- Collaborated with NASA Ames to test Astrobee and refine ISS deployment plans; stabilized the competition IDE through unit testing and 24 bug fixes, and served as instructor and Master of Ceremonies for the 2025 finals

University of Minnesota – Minneapolis, MN

NASA Space Grant Consortium Research Intern *Jun 2023 - Aug 2023*

- Developed a solar eclipse sun-tracking payload for a high-altitude balloon, enhancing software to improve reliability and achieve 360° tracking; programmed microcontrollers and integrated inertial, GPS, and atmospheric sensors for data collection and payload control

Engineering Projects & Teams

HackMIT — Cambridge, MA

Stratospheric Balloon Project | 24-Hour Hackathon *September 2026*

- Designed, built, launched, and recovered a stratospheric balloon payload carrying an NVIDIA Jetson within 24 hours, exchanging LLM prompts sent in by MIT students and responses over radio using a custom-built Yagi antenna
- Analyzed NOAA wind forecasts to identify Atlantic Ocean drift risk; evaluated doubling standard 2-lb free lift to accelerate ascent and reduce downrange drift, successfully recovering the payload within a 10-minute drive of MIT

MIT Rocket Team – Cambridge, MA

Liquid Propulsion Member | Former Solid Propulsion Member & Executive IT Chair *Sep 2024 - Present*

- Led FEA of a filament-wound CFRP motor case in Ansys/ACP- built the winding/laminate model, defined the orthotropic CFRP material, simulated internal-pressure and remote-displacement end loads, and correlated predictions to hydrostatic testing
- Machined nozzle carrier and forward closure, mixed/cast custom solid propellant, and designed grain geometry for low-altitude SRAD tests

FIRSTxMIT – Cambridge, MA

Head RI3D Captain *Oct 2025 - Present*

- Launched RI3D@MIT in its inaugural year, leading a ~25-student team through rapid CAD, design reviews, and software development; personally fronted ~$4,500 for time-critical parts and secured donated materials from local businesses, delivering a fully functional robot within 72 hours of kickoff

FIRST Robotics – Fairmont, MN

Captain, Programming Lead, and Driver of Team 2847 *Oct 2020 - Aug 2024*

- Led FRC Team 2847 as captain, programming lead, and robot driver, securing its first state berth, qualifying for world competition, and winning the Minnesota state championship
- Successfully organized $149,000 in fundraising through grant writing, community initiatives, and business searching, surpassing the program's total funds from the previous two decades

Technical Skills

- **Programming:** Python, Java, C++, C, GNU Make, Arduino, GCODE, Qt, SVG (markup)
- **Engineering & Simulation:** SolidWorks, Onshape, Ansys (FEA/ACP), Pointwise, US3D, Tecplot, openMotor, OpenRocket, RASAero
- **Hardware & Manufacturing:** NVIDIA Jetson/AGX, LiDAR, IMUs, GPS, CNC Machining, 3D Printing
- **Development Tools:** Git (CLI & GUI), GitHub, CANalyzer, WinMerge, Command Shell, PowerShell, Agile
- **Languages:** Spanish (Upper Intermediate)